HI-LO HIGHWAY

iGaming Card Prediction Game · Original Concept

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1. Executive Summary

Format: single-screen HTML5 game · mobile-first · card prediction with cash-out
RTP: 98.0%, exact for any strategy and any cash-out point
Genre: Hi-Lo / escalation, where the player picks the risk of every step
Board: 4 piles, each growing up to 5 cards (16 guesses max, 20-card full board)
Max win: ×10,000; any guess that would exceed it is disabled
Signature feature: four piles on the table and three calls (Higher, Same, Lower), so the player picks the risk of every step, from a near-safe ×1.06 to a 1-in-13 shot at ×12.74

2. Core Concept & Theme

2.1 Concept

Four cards are dealt face up, one per pile. On each turn the player picks a pile and calls Higher, Lower or Same. A new card lands on that pile: right call, the multiplier climbs; wrong call or same rank, the round is lost. The player can cash out at any time. Picking the pile is the decision that matters: as a first call, an Ace or a King is a near-safe step (×1.06), a 7 is a coin flip (×2.12), a Queen called Higher or any card called Same is a 1-in-13 shot (×12.74). The full board (20 cards) triggers an automatic cash-out.

2.2 Visual Direction

Kaching Originals house style, shared with Ricochet and Candy Monster: dark blue table (#0b1622), panels in #172533, green action button (#00e701), Inter font. Flat, high-contrast cards readable on a phone, with suits as decoration only. The four piles fan out vertically as they grow, so the board itself shows how far the run went.

2.3 Competitive Positioning

	Stake Hilo (Stake Originals)	Hi-Lo Highway
RTP	99%	98%
Cards in play	1 current card	4 persistent piles
Same rank	wins with "or same"	loses on Higher / Lower; separate Same call at ×12.74
Skip	yes	no; the pile choice replaces it
Run goal	open-ended	full board, 16 steps

3. Game Mechanics

3.1 Round Flow

- Before the bet, 4 cards lie face down. The player sets the bet and presses **Bet**: the stake is deducted and the board turns face up. Playtests showed players reach for the cards first, so tapping a face-down card makes the Bet button pulse and tells the player to bet first. The board is never visible before the stake is committed, so a player cannot refresh the page to shop for a board.
- Under each pile, three labelled buttons show the call (Higher in green, Same in gold, Lower in blue), the multiplier the player would reach and the chance of winning.

- The player taps one button. A new card is dealt onto that pile. Right call: the multiplier and the potential win update. Wrong call or same rank: the round is lost and the stake is gone.
- After the first right call, **Cash out** pays bet × multiplier. Each pile holds 5 cards; when all four piles are full, the win is cashed out automatically.
- After a cash-out or a loss, the result stays on screen for about 2 seconds, then the cards turn face down, ready for the next bet.

3.2 Rules

- Ace is low (1), King is high (13). Suits are cosmetic.
- Infinite shoe: every draw is uniform over the 13 ranks and independent of the cards on the board.
- On Higher or Lower, the same rank loses. "Higher" on a King and "Lower" on an Ace can never win, so these buttons are disabled.
- The **Same** call wins only when the next card has the same rank: 1 chance in 13 on any card, ×12.74 per step. It gives every pile a high-volatility option and turns the tie into something the player can aim for.
- A guess whose multiplier would exceed ×10,000 is disabled and shows MAX. If no guess remains, the win is cashed out automatically.

3.3 Interface

- Header: current multiplier (dimmed before the first right call, green on a win, red LOST on a loss) and a one-line status: potential win, cards found, reason for a loss.
- Piles fan downward; the newest card is in front. A dashed outline shows where the next card will land.
- Each guess button shows the resulting multiplier and the probability, so the player always sees the risk he is buying.
- Cash-out button shows the amount the player would collect.
- History of the last 16 rounds, session stats (rounds, cash-outs, best win), odds table per card under the game.
- Shortcuts: Space to bet, Enter to cash out. French and English.

4. Mathematical Model — Constant-RTP Ladder

4.1 Odds per Card

Chance of winning each call and the step factor (1 ÷ chance) applied to the multiplier:

Top card	Higher	Same	Lower
A	92.3% · ×1.08	7.7% · ×13.00	—
2	84.6% · ×1.18	7.7% · ×13.00	7.7% · ×13.00
3	76.9% · ×1.30	7.7% · ×13.00	15.4% · ×6.50
4	69.2% · ×1.44	7.7% · ×13.00	23.1% · ×4.33
5	61.5% · ×1.63	7.7% · ×13.00	30.8% · ×3.25
6	53.8% · ×1.86	7.7% · ×13.00	38.5% · ×2.60
7	46.2% · ×2.17	7.7% · ×13.00	46.2% · ×2.17
8	38.5% · ×2.60	7.7% · ×13.00	53.8% · ×1.86
9	30.8% · ×3.25	7.7% · ×13.00	61.5% · ×1.63
10	23.1% · ×4.33	7.7% · ×13.00	69.2% · ×1.44

Top card	Higher	Same	Lower
J	15.4% · ×6.50	7.7% · ×13.00	76.9% · ×1.30
Q	7.7% · ×13.00	7.7% · ×13.00	84.6% · ×1.18
K	—	7.7% · ×13.00	92.3% · ×1.08

4.2 Multiplier

After n right calls with winning chances $p_1 \dots p_n$, the multiplier is **$m = 0.98 \div (p_1 \times p_2 \times \dots \times p_n)$**. Each step multiplies the expected value by $p \times (1 \div p) = 1$, so the expected payout of any cash-out is 0.98 × the stake, whatever the piles chosen, the calls made or the moment of cash-out. The RTP is therefore exactly 98% for every strategy.

Example run on a 1 € stake:

Call	Chance	Step factor	Multiplier
Higher on Ace	92.3%	×1.08	×1.06
Lower on 7	46.2%	×2.17	×2.30
Higher on Queen	7.7%	×13.00	×29.90

Cashing out after these three calls pays 29.90 €. The chance of getting there was 3.28%.

4.3 Cap and Full Board

Guesses that would push the multiplier past ×10,000 are not offered, so the cap never cuts a payout and never lowers the RTP. The full board (16 right calls) and the "no guess left" case both cash out at the multiplier reached.

4.4 Simulation

Four strategies simulated on the same rules as the prototype, 16 million rounds in total (95% confidence intervals):

Strategy	Rounds	RTP
Prudent: safest call, cash out after 6	4,000,000	98.08% ± 0.22
Risky: longest shot, cash out after 2	4,000,000	97.94% ± 0.77
Random call, cash out after 4	4,000,000	97.18% ± 0.93
Full board: safest call every time	4,000,000	97.97% ± 2.65

Every result is consistent with 98%. Wide intervals come from rare very large multipliers, which is also why the RTP is proven analytically (4.2) rather than only measured.

5. Audio Design

- Card deal: short filtered noise flick.
- Right call: two-note triangle chime that rises in pitch with the length of the run.
- Wrong call: falling sawtooth tone.
- Cash-out: four-note ascending arpeggio.
- All sounds synthesised with the Web Audio API; Sound ON / OFF toggle.

6. Technical Specifications

- One HTML file: DOM cards and CSS animations (deal, win glow, loss shake), vanilla JavaScript, no image assets.
- RNG: crypto.getRandomValues; each card is drawn at the moment of the call and shown immediately.
- Responsive layout from 320 px phones to desktop; card size computed from the board width.
- French and English interface; play-money balance shared with the other portfolio games.

7. Commercial Roadmap

- RGS integration with server-side outcome generation and a certified RNG.
- Math certification (GLI-19 / BMM) of the ladder.
- RTP variants by changing the 0.98 factor (for example 97% or 96%).
- Finite-shoe variant (6 decks, cards removed as they are dealt), which turns the board into a light card-counting puzzle and needs its own pricing.